RESEARCH PAPER / OCTOBER 2026

Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor | Case-study dataset

10,194	8,549	196	5
SKU line items	Dispatch records	Factory-state pairs	Assumed factories

Critical evidence warning. The source ship dates (2026-2030) are inconsistent with 2024-2025 orders. All raw intervals exceed one year. Any adjusted metric presented below is unverified exploratory sensitivity, not measured company delivery performance.

Abstract

This paper implements a reproducible shipping analytics prototype from 10,194 order line items and 8,549 distinct factory-to-destination dispatches. Modules include order/date validation, case-study product-to-factory assignment, volume-screened route benchmarking, a US geographic view, mode comparison and order-level exploration. The primary research finding is that the original dates are unsuitable for an ordinary logistics-time claim.

Three disjoint 365-day-spaced interval clusters motivate a clearly unverified diagnostic adjustment. Its mean is 4.29 days, compared with the raw 1321.37 days. This difference is an assumed data-repair sensitivity rather than confirmed shipping performance. Source lineage and actual delivery events are prerequisites for production decisions.

Core research questions

Which assumed factory-to-state pairs display relatively longer intervals? How do supplied shipping service categories compare? Which high-volume states merit further examination? What additional data are necessary to support verified transit reliability and freight-cost decisions?

Evidence class: student/educational case study. Factory names and coordinates come from the project brief, not independently verified Nassau Candy facilities.

1 Source data and integrity audit

Each row represents a SKU line; some order IDs repeat across lines. The analysis consolidates identical order-factory-destination-dispatch combinations. The source file has 18 columns, 15 product names, four shipping modes and both US and Canadian destinations.

Validation check	Observed result
SKU-level rows	10,194
Distinct dispatch records	8,549
Invalid/missing date rows	0
Negative lead-time rows	0
Unmapped factory product rows	0
Dispatches with raw lead over 365 days	8,549
Future ship dates at 2026-10-12	8,245
US / Canada dispatch records	8,389 / 160

The dates span orders from 2024-01-02 to 2025-12-31; ship dates from 2026-06-30 to 2030-06-28. Source order-to-ship intervals range from 904 to 1,642 days. These values cannot credibly be interpreted as ordinary logistics fulfillment or delivery time.

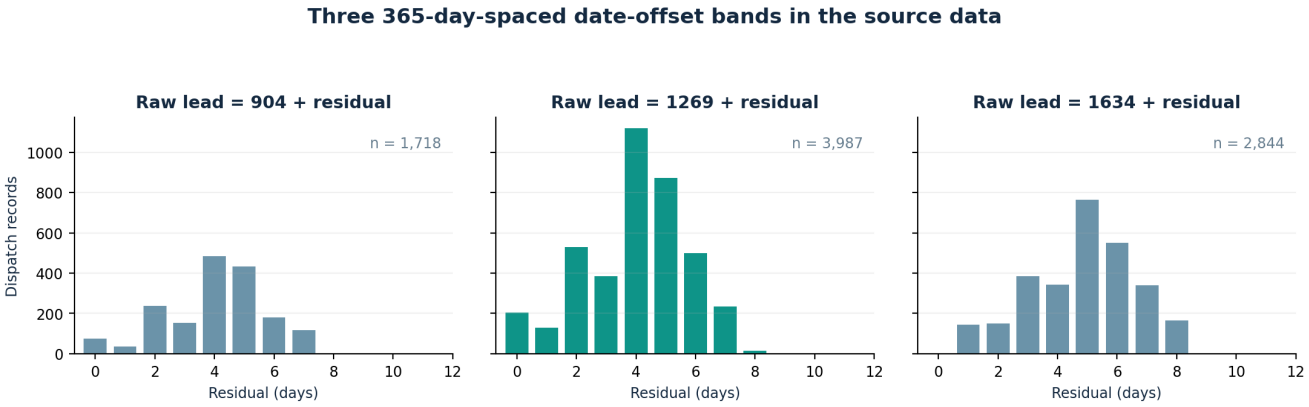


Figure 1. Three discrete date-offset groups separated by 365 days. The x-axis is the remainder after subtracting each group base; this visual does not validate the proposed correction.

Integrity finding

Intervals form three distinct groups with bases at 904, 1,269 and 1,634 days. The second and third bases differ by exactly 365 days. Their existence is compatible with artificial date shifts, but the source mechanism and correct timestamps are unknown. Original raw dates are therefore preserved in the app.

2 Cleaning and analytical design

Date fields are parsed explicitly as day-month-year, geographic labels are trimmed and standardized, and product names map to the 15 assignment-defined factory lookups. Rows with missing or invalid timestamps, negative intervals, or unknown factories are excluded from performance calculations but counted in the validation audit. Multi-line SKU purchases are consolidated at order-factory-destination-dispatch grain.

KPI definitions

Metric	Operational definition
Raw lead time	Calendar-day difference: Ship Date minus Order Date. This is NOT carrier delivery duration.
Diagnostic residual	For raw lead in one of the three 0-14-day offset bands, subtract 904, 1269 or 1634 days. Unverified assumption.
Route volume	Number of distinct dispatch records for the assigned factory and customer state/province.
Variability	Standard deviation of dispatch intervals, with 90th percentile shown separately.
Delay percentage	Fraction of dispatch intervals strictly above an adjustable five-day diagnostic threshold. Not an SLA.
Efficiency score	Within eligible routes: 100 x (max mean - route mean) / (max mean - min mean); ties score 100. Relative only.

Research screening and bottleneck rule

The report requires at least 50 dispatch records per factory-to-state route (34 routes qualify), while the dashboard minimum defaults to 20 and remains adjustable. The benchmark is a descriptive ranking, not an estimate of miles driven, fuel use or monetary cost. A state is flagged as a candidate only when its dispatch volume meets or exceeds the median state volume AND its average exceeds the overall average.

Scenario design / caution

The default interactive dashboard opens in an explicitly labeled exploratory adjusted mode so users can examine residual patterns; raw intervals are available one click away. The adjustment subtracts suspected artificial offsets only when the remainder lies between zero and fourteen days. Values outside that band are not silently repaired. No mode-to-customer service guarantee can be inferred without verified dates.

Geographic treatment

The US choropleth uses state codes and excludes Canada, which is summarized separately. Optional straight connections from supplied factory coordinates to approximate state centroids are illustrations and are NOT actual freight or road itineraries. The case-study factories are not documented actual Nassau Candy origins.

3 Service-mode and regional results

Interpretation condition. All adjusted results on this page are diagnostic sensitivity values and should NOT be cited as Nassau Candy verified transport or delivery KPIs.

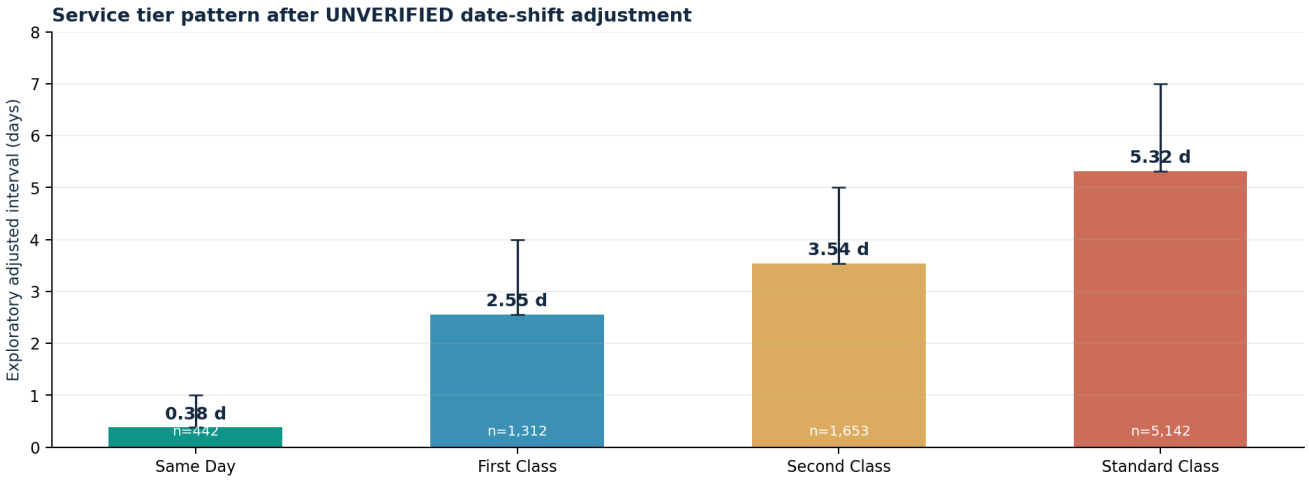


Figure 2. Mean diagnostic residuals by service mode; upper whisker reaches each mode's 90th percentile.

Ship mode	Dispatches	Mean days	Over 5 days
Same Day	442	0.38	0.0%
First Class	1,312	2.55	0.0%
Second Class	1,653	3.54	6.7%
Standard Class	5,142	5.32	38.8%

All-route diagnostic mean

Across 8,549 dispatches, adjusted mean is 4.29 days, median 4.0, 90th percentile 7.0, and the fraction exceeding five days is 24.6%. The mode ordering resembles expected service tiers but does not certify the source-date adjustment.

Region-level aggregates

Customer region	Dispatches	Avg days	Over 5 days
Atlantic	2,476	4.26	23.5%
Pacific	2,744	4.27	25.2%
Gulf	1,350	4.29	24.0%
Interior	1,979	4.38	25.5%

Regional differences are modest relative to the interval scale, and uncontrolled differences in mode mix, factory, season or customer mix may account for apparent variation.

4 Route benchmarking

These two ranked tables use the exploratory residual and exclude routes with fewer than 50 distinct dispatches. Raw source lead-time ranks are available through the dashboard but are heavily confounded by the multi-year date-shift artifact.

Top 10 lower-interval eligible routes

#	Factory to destination	N	Mean d	>5 d
1	Lot's O' Nuts -> Ohio	218	3.90	23.4%
2	Lot's O' Nuts -> Virginia	91	3.91	14.3%
3	Wicked Choccy's -> Ohio	157	3.96	19.7%
4	Lot's O' Nuts -> Massachusetts	58	4.00	13.8%
5	Lot's O' Nuts -> Colorado	82	4.05	22.0%
6	Lot's O' Nuts -> California	944	4.12	24.0%
7	Wicked Choccy's -> Georgia	55	4.13	21.8%
8	Lot's O' Nuts -> Pennsylvania	270	4.14	16.3%
9	Wicked Choccy's -> Virginia	85	4.19	20.0%
10	Lot's O' Nuts -> Michigan	118	4.20	26.3%

Bottom 10 higher-interval eligible routes

#	Factory to destination	N	Mean d	>5 d
1	Lot's O' Nuts -> New Jersey	53	4.92	41.5%
2	Wicked Choccy's -> Tennessee	56	4.84	32.1%
3	Lot's O' Nuts -> Tennessee	88	4.73	33.0%
4	Wicked Choccy's -> Michigan	83	4.59	34.9%
5	Wicked Choccy's -> Illinois	183	4.56	27.9%
6	Lot's O' Nuts -> Illinois	235	4.52	23.0%
7	Lot's O' Nuts -> Indiana	74	4.47	28.4%
8	Lot's O' Nuts -> Oregon	62	4.45	32.3%
9	Lot's O' Nuts -> Arizona	94	4.41	29.8%
10	Lot's O' Nuts -> Washington	230	4.38	25.2%

The lowest and highest eligible routes should be treated as investigation candidates only. Differences may arise from service mix or date-cohort composition, not true geographic inefficiency. This is a descriptive screen, not causal evidence or optimized fleet routing.

5 Geographic signals and throughput

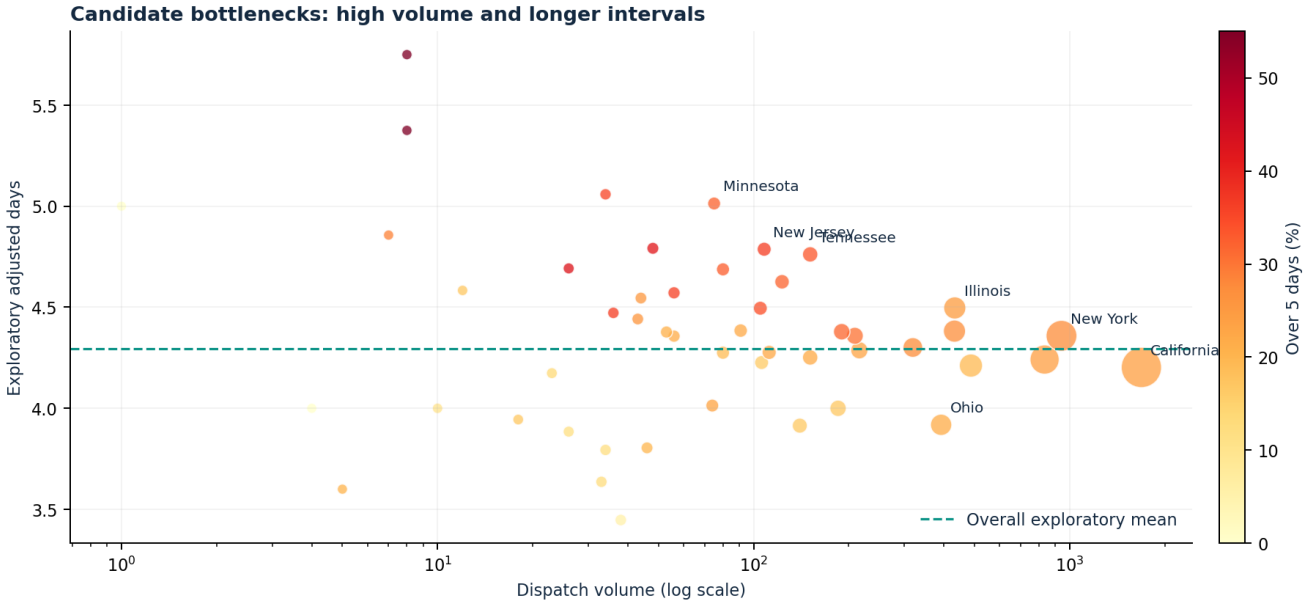


Figure 3. Each bubble is a US destination state; volume is on the x-axis (log scale), mean diagnostic residual on y-axis, with color showing fraction over five days.

Higher-volume, higher-interval candidates

State / province	Dispatches	Avg days	>5 days
Arkansas	48	4.79	41.7%
New Jersey	108	4.79	36.1%
Oklahoma	56	4.57	35.7%
Oregon	105	4.50	34.3%
Tennessee	151	4.76	33.1%
Delaware	80	4.69	32.5%
Minnesota	75	5.01	32.0%
Indiana	123	4.63	31.7%

A state appearing in this list is not proof of network congestion. Before investigating routes or carriers, require real delivery events, route distances, origin site IDs, mode adjustment and source-date reconciliation. Canadian province records are kept in separate tables rather than projected onto US state polygons.

6 Time trends, commercial variables and gaps

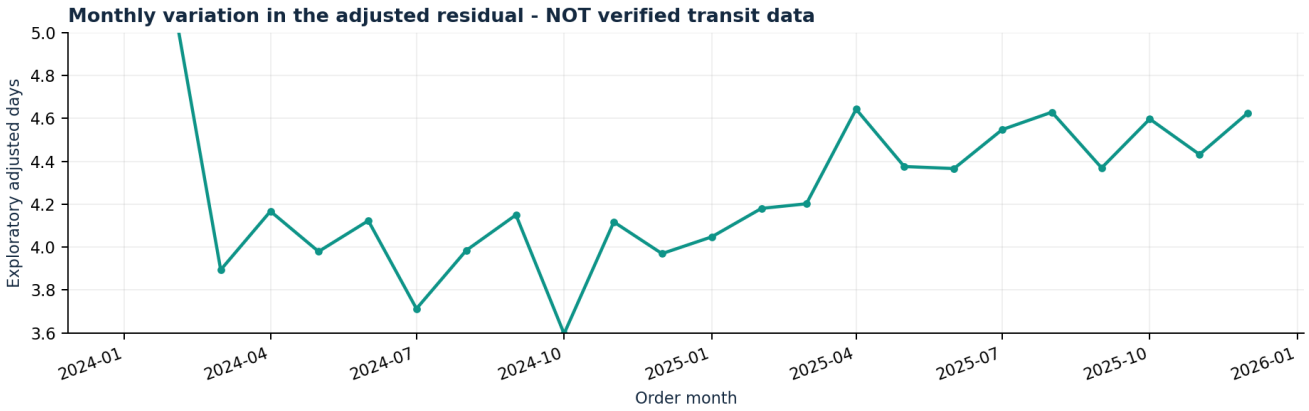


Figure 4. Diagnostic scenario mean by original order month; only shows sensitivity to the inferred correction.

No validated freight cost / time frontier

The dataset financial columns contain Sales, Units, Cost and Gross Profit. Cost is explicitly manufacturing cost and Gross Profit = Sales - Cost. Neither field provides carrier charges, transportation cost, fuel, miles, weight, route path, actual delivery time or promised delivery date. Consequently comparing these financial columns with adjusted lead time would NOT estimate the monetary value of expedited delivery.

Most important methodological limitations

First, shipment dates appear altered or synthetic. Second, the date difference is order-to-ship, not shipping-to-delivery. Third, factory labels and coordinates come from the assignment, not corporate records. Fourth, the sample overweights chocolate products relative to other categories, and the number of dispatches differs widely between factories. Fifth, all rankings are unadjusted for product, service class, calendar cohort, route distance and regional seasonality. The sample includes Canada, whose geographic representation must be kept distinct from a US-state map.

Company-site fact check

Nassau Candy public pages describe a national wholesale distribution business with warehouses in New York, Michigan, Florida, Texas and California. The project brief instead specifies five Wonka-named factories; this paper does not claim those names or geographic points are actual Nassau Candy facilities.

7 Recommendations, conclusion and sources

Priority 1 | Reconcile data before decisions

Obtain original ERP/WMS order, fulfillment and shipment event logs; carrier manifest and proof-of-delivery scans; time-zone and date-format definitions; and a sample of source transactions. Validate or reject the 904/1269/1634-day offset inference. Do not publish raw or adjusted intervals as official performance in the meantime.

Priority 2 | Acquire the missing operational attributes

Add actual ship-to-delivery dates, scheduled vs actual pickup, promised delivery deadline, carrier, true origin facility identifier, destination geocode, mileage, payload and freight spend. Split KPI families into fulfillment lead time, in-transit lead time, on-time delivery and freight cost per order / unit.

Priority 3 | Pilot accountable monitoring

After reconciliation, report weekly distributions with minimum-volume qualifications, service-mix controls and driver-level exception labels. Use states with meaningful volumes and repeat anomalies as candidates for investigation, not automatic site or vendor rankings. Govern Canadian and US operational jurisdictions separately.

Conclusion

The project meets the analytical prototyping objective while exposing a decisive evidence limitation. The exploratory 4.29-day adjusted mean may be useful as a test scenario but not as an actual performance fact. A verified carrier/delivery dataset can later unlock trustworthy geographic bottleneck detection, shipping-cost comparison and route service optimization.

Sources and provenance

1. Assignment-supplied Nassau Candy CSV (18 columns), factory-location coordinates and product-to-factory lookup; provenance not independently verified.

2. Nassau Candy official home: https://www.nassaucandy.com/ (accessed 12 October 2026).

3. Nassau Candy delivery information: https://www.nassaucandy.com/delivery-information (accessed 12 October 2026).

4. Nassau Candy contact / distribution centers: https://www.nassaucandy.com/contact/ (accessed 12 October 2026).

Reproduce analyses: install requirements and run python scripts/generate_reports.py; tests: python -m pytest -q; dashboard: streamlit run app.py.